

What is deep learning



Artificial Intelligence

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What is deep learning

Deep learning is a subset of [machine neural networks](#), called deep neural networks, which harness the decision-making power of the human brain. It powers most of the [artificial intelligence](#) applications we use today.

The chief difference between deep learning and traditional machine learning is the underlying neural network architecture. “Non-linear” models use simple neural networks with one or two layers, while deep learning models use three or more layers, but typically hundreds of layers.

While [supervised learning](#) models require structured inputs and outputs, deep learning models can use [unsupervised learning](#). In unsupervised learning, deep learning models can extract the features they need to make accurate outputs from raw data. Some deep learning models can even evaluate and refine their own outputs.

Deep learning is an aspect of data science that aims to improve [automation](#), performing analytical and predictive tasks without human intervention. This enables many everyday products, such as virtual assistants, voice-enabled TV remotes, credit card fraud detection, and generative AI.

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How deep learning works

Neural networks, or artificial neural networks, are a combination of data inputs, weights and bias elements work together to accurately recognize data.

Deep neural networks consist of multiple layers. The previous layer to refine and optimize the output. The process of computations through the network is called forward pass. The layers of a deep neural network are called *vis*. The learning model ingests the data for processing. When a prediction or classification is made.

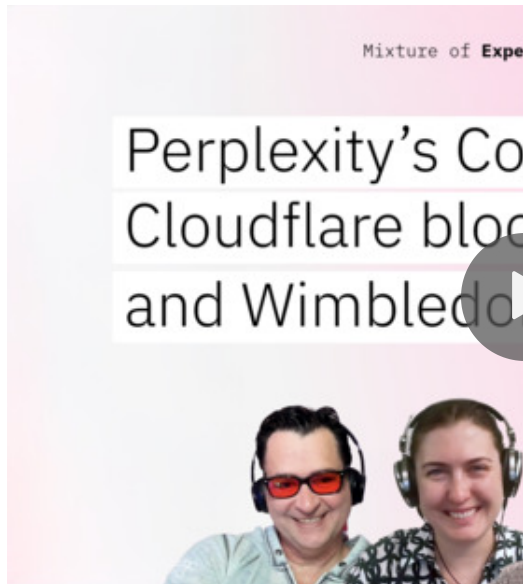
Another process called backpropagation uses the forward pass to calculate errors in predictions, and then adjusts the weights by moving backwards through the layers to train the model. Forward and backpropagation enable a neural network to learn from its errors. Over time, the algorithm becomes more accurate.

Deep learning requires a tremendous amount of computational performance [graphical processing units \(GPU\)](#) to handle the volume of calculations in multiple cores with

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computing might also assist. This level of complexity is achieved through deep learning algorithms. However, running these algorithms can create a large demand on internal resources and infrastructure requirements, most deep learning apps are created using frameworks: JAX, PyTorch or TensorFlow.

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Types of deep learning

Deep learning algorithms are incredibly complex neural networks to address specific problems or data sets. They have several advantages and they are presented here roughly in order of increasing complexity, each successive model adjusting to overcome the limitations of the previous one.

One potential weakness across them all is that they are often seen as "black boxes," making it difficult to understand their internal workings.

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challenges. But this can be balanced against the scalability.

CNNs

Convolutional neural networks (CNNs or ConvNets) are used for image and image classification applications. They can process images and videos, enabling tasks such as object recognition and face recognition. These networks use, particularly matrix multiplication, to identify patterns in data.

CNNs are a specific type of **neural network**, with an input layer, one or more hidden layers and another and has an associated weight and threshold. If the value is above the specified threshold value, that node is passed to the next layer of the network. Otherwise, no data is passed.

At least three main types of layers make up a CNN: the input layer, the hidden layer, and the output layer.

Each layer builds on the previous layer. The input layer takes the original input detailed patterns can be discovered in its complexity, identifying greater portions of the input. The hidden layer identifies features, such as colors and edges. As the image moves through the CNN, it starts to recognize larger elements or the intended object.

CNNs are distinguished from other neural networks by their ability to process image, speech or audio signal inputs. Before CNNs, feature extraction methods were used to identify objects in images. CNNs are a more scalable approach to image classification of high-dimensional data. And CNNs can exchange information more efficiently than other methods. While information can be outweighed by the benefits of CNNs, which can improve efficiency and limit risk of overfitting.

There are other disadvantages to CNNs, which include the time and budget, requiring many graphical processing units (GPUs) and highly trained experts with cross-domain knowledge to tune hyperparameters and configurations.

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RNNs

Recurrent neural networks (RNNs) are typical recognition applications as they use sequentially identified by their feedback loops. These learn using time-series data to make predictions at stock market predictions or sales forecasting, language translation, natural language processing, captioning. These functions are often incorporated into voice search and Google Translate.

RNNs use their “memory” as they take information from current input and output. While traditional deep learning outputs are independent of each other, the outputs are dependent within the sequence. While future events would be predicted from a given sequence, unidirectional recurrent neural networks focus on events in their predictions.

RNNs share parameters across each layer of the network. Each parameter within each layer of the network, via the processes of backpropagation and gradient descent, is updated.

RNNs use a backpropagation through time (BPTT) which is slightly different from traditional backpropagation. The principles of BPTT are the same as traditional backpropagation. As the model trains itself by calculating errors from inputs and outputs, BPTT differs from the traditional approach in that BPTT sums errors across all layers of the network. Feedforward networks do not need to sum errors across layers.

An advantage over other neural network types is their ability to process and store information. RNNs can plan out next steps rather than delivering only one result for a single input. They can handle one-to-one or many-to-many outputs.

There are also options within RNNs. For example, a Long Short-Term Memory (LSTM) network is superior to simple RNNs by learning long-term dependencies.

However, RNNs tend to run into two basic problems: vanishing gradients. These issues are defined by the vanishing gradient problem, where the gradient of the loss with respect to the input vanishes as it propagates back through the network.

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slope of the loss function along the error curve

- When the gradient is *vanishing* and is too small, updating the weight parameters until they reach a minimum that occurs, the algorithm is no longer learning.
- *Exploding* gradients occur when the gradient becomes too large. In this case, the model weights grow too large and become NaN (not a number). One solution to this problem is to use layers within the neural network, eliminating models.

Some final disadvantages: RNNs might also require a lot of memory use on large datasets. Optimizing RNNs adds a lot of parameters.

Autoencoders and variational autoencoders

Deep learning made it possible to move beyond the analysis of images, speech and other common tasks. Models to achieve this were [variational autoencoders](#). These learning models to be widely used for generative modeling by making them a cornerstone of what we think of as [generative AI](#).

Autoencoders work by encoding unlabeled data into a latent space and then decoding the data back into its original form. They have a variety of purposes, including reconstructing data. Variational autoencoders added the critical ability not just to reconstruct but also to generate variations on the original data.

This ability to generate novel data ignited a race to create more realistic but fake images. In this way, generative AI.

Autoencoders are built out of blocks of encoding and decoding. The encoding underpins today's large language models. Encoders create a representation, arranging similar data points into a latent space. Decoders sample from this space to create some of the most important features.

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The biggest advantage to autoencoders is the show input data in a compressed form, so the anomaly detection and classification tasks. TI storage requirements. Autoencoders can be t used where labeled data is not available. Whe time savings advantage: deep learning algorit without needing manual feature engineering. data for text or image generation.

There are disadvantages to autoencoders. Th be a drain on computational resources. And d might overlook the needed properties and ins Autoencoders might also overlook complex d not correctly identify complex relationships.

GANs

Generative adversarial networks (GANs) are r outside of artificial intelligence (AI) to create data. These can include images appearing to of real people. The “adversarial” part of the n between the two portions of the GAN: a gene

- The **generator** creates *something*: images, output with a twist. For example, a horse c degree of accuracy. The result depends or are in the generative model for this use ca
- The **discriminator** is the adversary, where compared against the *real* images in the d between the real and fake images, video c

GANs train themselves. The generator create the differences between the generator's fake discriminator is able to flag the fake, then the continues until the generator succeeds in pro distinguish.

The prime GAN benefit is creating realistic ou the originals, which in turn may be used to fur up a GAN to learn is straightforward, since the minor labeling. However, the potential disadv

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discriminator might go back-and-forth in computer system drain. One training limitation is that a generator produces a limited set of outputs rather than a continuous range.

Diffusion models

Diffusion models are generative models that learn the diffusion process of progressive noise-addition to data, most often images similar to the data or the data used to train them. They gradually add noise until it's unrecognizable, then learn a reversed "denoising" process to generate (usually images) from random noise input.

A diffusion model learns to minimize the difference between the generated output and the desired target. Any discrepancy is quantified as a loss, and the model is trained to minimize the loss by adjusting its parameters.

Beyond image quality, diffusion models have several advantages over GANs, including faster training, which speeds the learning process and reduces the risk of mode collapse. Training is more stable than with GANs and does not require adversarial training.

But, compared to GANs, diffusion models can be more computationally intensive, especially when including more fine-tuning. IBM Research® has found that generative AI can be hijacked with **hidden backdoors** to manipulate the image creation process so that AI diffusion models generate manipulated images.

Transformer models

Transformer models combine an encoder-decoder architecture and have revolutionized how language models process text. They convert raw, unannotated text into representations and take these embeddings together with previous context to predict each word in a sentence.

Using fill-in-the-blank guessing, the **encoder** processes input tokens, and the **decoder** generates output tokens, building up a powerful representation of the input sequence.

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parts of speech and other grammatical features at the outset without a particular task in mind. Once learned, the models can later be specialized for a task.

Several innovations make this possible. Transformers simultaneously, enabling text processing in parallel. Techniques including recurrent neural networks and Transformers also learned the positions of words, which enables them to infer meaning and disambiguate.

By eliminating the need to define a task upfront, language models on vast amounts of raw text. Previously, labeled data was gathered to train transformers, one model trained on a massive dataset of tasks by fine-tuning it on a small amount of labeled data.

Language transformers today are used for natural language processing tasks such as entity extraction as well as generative tasks like text completion and question answering. Transformers have also been used to generate convincing dialog, essays and other content.

Natural language processing (NLP) transformers run in parallel, processing multiple portions of text, which greatly speeds training. Transformers also train on large datasets, which enables them to understand the overall context. In addition, transformers are more scalable and efficient.

As to limitations, because of their complexity, they require significant resources and a long training time. Also, the models are not unbiased and plentiful to produce accurate results.

Deep learning use cases

The number of uses for deep learning grows exponentially. It is now helping businesses become more efficient and improve customer experiences.

Application modernization

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Generative AI can enhance the capabilities of skills gap in the domains of [application mode for coding](#) is possible because of recent breakthrough technologies and natural language processing. Large neural networks trained on vast dataset generally comes from publicly available code.

Programmers can enter [plain text prompts](#) and Generative AI tools suggest code snippets or process by handling repetitive tasks and reduce. Translate code from one language to another, modernization projects, such as updating legacy Java.

Computer vision

[Computer vision](#) is a field of artificial intelligence for object detection and semantic segmentation. It uses deep learning networks to teach computers and learning systems to process digital images, videos and other visual inputs. It takes actions when the system sees defects or issues. Computer vision enables them to see, observe and analyze.

Because a computer vision system is often trained on production assets, it usually can analyze thousands of images, noticing imperceptible defects or issues. Companies from energy and utilities to manufacturing and healthcare use it.

Computer vision needs lots of data, and then it learns until it discerns and ultimately recognizes images. To recognize automobile tires, it needs to be fed thousands of related items to learn the differences and recognize them.

Computer vision uses algorithmic models to extract the context of visual data. If enough data is fed through it, it can learn at the data and teach itself to tell one image from another to learn by itself, rather than with someone providing labels.

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Computer vision enables systems to derive meaning from images, videos and other visual inputs, and based on this, it can [provide recommendations](#) that distinguishes it from common applications of computer vision today.

- **Automotive:** While the age of driverless cars, self-driving technology has started to make its way into the market, passenger safety through features such as lane departure warnings.
- **Healthcare:** Computer vision has been increasingly used, enabling doctors to better identify cancerous cells in medical images.
- **Marketing:** Social media platforms provide personalized recommendations of a photograph that has been posted on a person's social media albums.
- **Retail:** Visual search has been incorporated into e-commerce, enabling brands to recommend items that are similar to those a customer is viewing.

Customer care

AI is helping [businesses](#) to better understand their customers. With the rise of highly personalized online shopping and delivery services, generative AI can help further enhance customer care, talent transformation and the overall customer experience.

AI empowers businesses to adopt a customer-centric approach by using insights from customer feedback and buying behavior to improve product design and packaging and create personalized offers that increase sales.

Generative AI can also serve as a cognitive assistant, providing contextual guidance based on conversation history and transcripts. Also, generative AI can enable personalized customer loyalty and provide a competitive advantage.

Digital labor

Organizations can augment their workforce by using robotic process automation (RPA) and [digital labor](#) to collaborate with human workers.

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assist whenever backup is needed. For example, updating of legacy software.

Digital labor uses foundation models to automate knowledge workers by enabling self-service and removing technical barriers. To automate task performance, an LLM-based slot filling model can identify information required for completing an action with minimal effort.

Instead of having technical experts record and train knowledge workers, digital labor automations use conversational instructions and demonstrations to create self-service automation. For example, to speed up development, it can help end-users, who lack programming experience, with writing and validating code.

Generative AI

Generative AI (also called gen AI) is a category of AI that generates images, video, data or other content in response to a prompt.

Generative AI relies on deep learning models trained on large amounts of content and generate new, similar content based on prompts. In many fields including customer service, marketing, and design, it offers enormous potential to streamline enterprise operations, automate content creation and augmentation.

Generative AI excels at handling diverse data types, such as audio files and social media content. This unlocks new ways of creating models and the ongoing training of generative models over time. Using this unstructured data can enhance customer service, facilitate more effective email routing. In practice, it can connect appropriate resources, whether that's connecting a customer to user guides and FAQs.

Despite its much-discussed limitations and risks, organizations are cautiously exploring how their organizations can leverage generative AI to internal workflows, and enhance their productivity. Generative AI to make the workplace more efficient without

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AI agents and agentic AI

An **AI agent** is an autonomous AI program—it behaves on behalf of a user or another system without human intervention, following a workflow and using available tools (other applications or APIs).

Agentic AI is a system of multiple AI agents, typically working together and orchestrated, to accomplish a more complex task than any one agent in the system could accomplish.

Unlike chatbots and other AI models which only respond to prompts, AI agents and agentic AI exhibit autonomous behavior and adaptability to changing circumstances. We often refer to these models' *agency*, or their capacity to act on their own.

One way to think of agents is as a natural next step in the evolution of AI, moving from a focus on creating content based on learned patterns to interacting with each other and other tools to make decisions. For example, a generative AI app might be able to tell you the weather given your work schedule, but an agent can tell you the weather and offer service to book you the best flight and reservation for your trip to Nepal.

[Explore our 2025 guide to AI agents](#) →

Natural language processing and recognition

NLP combines computational linguistics, rule-based and statistical models, and machine learning models to enable systems to recognize, understand and generate text and speech. NLP devices that can translate text from one language to another, execute commands, recognize or authenticate users by voice, process volumes of text, assess the intent or sentiment of text, or generate graphics or other content on demand.

A subset of NLP is statistical NLP, which combines machine learning and deep learning models. This approach uses large amounts of data to train models to recognize patterns in text and speech.

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and label elements of text and voice data and possible meaning of those elements. Today, techniques based on RNNs enable NLP systems more accurate meaning from huge volumes of voice datasets.

Speech recognition: also known as automatic recognition or speech-to-text is a capability to convert speech into a written format.

While speech recognition is commonly confused with voice recognition, speech recognition focuses on the translation of speech into text, whereas voice recognition just seeks to identify the speaker.

Industry application

Real-world deep learning applications are all products and services that users are unaware are taking place in the background. Some of these include:

Customer service deep learning

Many organizations incorporate deep learning into their **customer service** processes. **Chatbots** are often used in service portals. Traditional chatbots use natural language processing (NLP) rules commonly found in call center-like menus. **AI chatbots** attempt to determine, through learning, the intent of ambiguous questions in real time. Based on this, they try to answer these questions directly or route them to a human agent.

Virtual assistants such as Apple's Siri, Amazon's Alexa, and Google Assistant are examples of a chatbot by enabling speech recognition functionality to engage users in a personalized way.

Financial services analytics

Financial institutions regularly use **predictive analytics** to analyze stock markets, assess business risks for loan approvals, and more.

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investment portfolios for clients.

Healthcare record-keepi

The [healthcare](#) industry has benefited greatly the digitization of hospital records and image support medical imaging specialists and radiologists to view more images in less time.

Law enforcement uses d

Deep learning algorithms can analyze and learn dangerous patterns that indicate possible fraud. Facial recognition, computer vision and other deep learning techniques increase the efficiency and effectiveness of investigative analysis from sound and video recordings, images and [enforcement](#) analyze large amounts of data from

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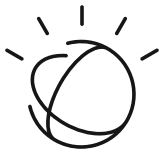
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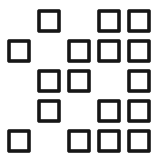
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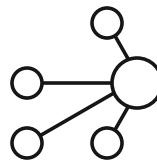
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